PRINCIPAL PARTS OF ORDER K AS A QUOTIENT OF THE BASE CHANGE BY THE POWER OF THE DIAGONAL IDEAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted proof ends by leaving the verification of the inverse identity to the reader; that sentence is kept verbatim inside the span and both maps are written out.

Editorial note. The counted proof refers to a same-chapter statement by label; it is recorded as a navigation stub pointed at its \label line.

Allowed standard prerequisites (editorial).

- The universal differential operator of order k. Used throughout the counted proof (lines 35160-35184). Supplies the map that the isomorphism has to identify with the class of one tensor m.
- The diagonal ideal as the kernel of the multiplication map. Used in the statement and in the counted proof. Provides the ideal whose (k+1)-st power cuts out the target of the isomorphism.

Source reference. Original lemma 09CK: sources/stacks-algebra/pinned-original-source.tex, source lines 34918–34918. Navigation pointer for \ref{lemma-module-principal-parts} used inside the counted span (line 35172); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $R \to S$ be a ring map. Let M be an S-module. Let $J = \mathrm{Ker}(S \otimes_R S \to S)$ be the kernel of the multiplication map. There is a canonical isomorphism of S-modules*

$$P^k_{S/R}(M) \longrightarrow (S \otimes_R M)/J^{k+1}(S \otimes_R M)$$

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where $s \in S$ acts on the target via multiplication by $s \otimes 1$ and such that the universal differential operators of order k to the map given by $m \mapsto$ class of $1 \otimes m$.

Proof. Consider the map $T : M \to S \otimes M$, $m \mapsto 1 \otimes m$ Since T is R-linear and since

$$g_0 g_1 \ldots g_k T(m) - \sum g_0 \ldots \hat{g}_i \ldots g_k T(g_i m) + \ldots + (-1)^{k+1} T(g_0 \ldots g_k m) = \prod_{i=0,\ldots,k} (g_i \otimes 1 - 1 \otimes g_i) 1 \otimes m$$

for $m \in M$ and $g_0, \ldots, g_k \in S$, we conclude that the rule $m \mapsto$ class of $1 \otimes m$ in the statement of the lemma is a differential operator of order k (see proof of Lemma 09CK). By the universal property of $P^k_{S/R}(M)$ we obtain the arrow in the statement of the lemma. On the other hand, if $D : M \to N$ is a differential operator of order k, then we can consider the S-linear map $L : S \otimes_R M \to N$, $g \otimes m \mapsto gD(m)$. The reader checks, by a computation similar to the one above and using that J is generated by elements of the form $g \otimes 1 - 1 \otimes g$, that L annihilates $J^{k+1}(S \otimes_R M)$. In particular, if we apply this to the universal differential operator $D_{univ} : M \to P^k_{S/R}(M)$, then we obtain an S-linear map $(S \otimes_R M)/J^{k+1}(S \otimes_R M) \to P^k_{S/R}(M)$. We leave it to the reader to see that this map is the inverse to the map in the statement of the lemma. $\square$